

Jeehong Kim

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RESEARCH INTEREST

I am interested in designing broadly applicable machine learning techniques for graph-based problems, especially in heterophilous and anomalous scenarios.

- **Keywords:** Graph Representation Learning, Graph Heterophily, Anomaly Detection, GNNs with LLM

EDUCATION

Seoul National University, Seoul, South Korea

- M.S.-Ph.D. Integrated Program in Data Science Mar 2023 – Present
 - Research Interest: Graph Representation Learning, Graph Heterophily, Anomaly Detection
 - Advisor: Prof. Hyunwoo Park
- Planning to fulfill military service as a *Technical Research Personnel* starting around September 2026

Korea University, Seoul, South Korea

- B.S. in Industrial Management Engineering Mar 2019 – Feb 2023
 - GPA: 3.76 / 4.50

RESEARCH EXPERIENCE

DSAIL (Data Science & Artificial Intelligence), KAIST

- Undergraduate Research Intern Jun 2022 – Aug 2022
 - Research Interest: Graph Neural Network, Recommendation System
 - Advisor: Prof. Chanyoung Park

AIDA Lab (Artificial Intelligence & Data Analytics), Korea University

- Undergraduate Research Intern May 2021 – Jan 2022
 - Research Interest: Anomaly Detection, GAN
 - Advisor: Prof. Sungwon Han

PUBLICATIONS

(†: Equal contribution)

JOURNALS

- [J1] FURIOUS: Fully unified risk-assessment with interactive operational user system for vessels
Yooyeun Kim[†], **Jeehong Kim[†]**, Wonhee Lee, Hyunwoo Park, Deuk Jae Cho
PLoS One (2025.05)
- [J2] STGVAD: Spatio-Temporal Graph-based Vessel Behavior Anomaly Detection
Jeehong Kim[†], Minchan Kim[†], Youngseok Hwang, Sungho Bae, Deuk Jae Cho, Wonhee Lee, Hyunwoo Park
IEEE Access (2025.09)

WORKSHOPS

- [W1] Adaptive Sparsified Graph Learning Framework for Vessel Behavior Anomalies
Jeehong Kim[†], Minchan Kim[†], Jaeseong Ju, Youngseok Hwang, Wonhee Lee, Hyunwoo Park
AAAI 2025 Workshop in Anomaly Detection in Scientific Domains

PROJECTS

MOIN

- Fintech Technology Commercialization Support Program Jul 2025 – Present
 - Fraud detection system (FDS) modeling optimization for overseas remittance using AI technology

Korea Research Institute of Ships & Ocean Engineering (KRISO)

- Vessel Behavior Anomaly Detection Oct 2024 – Present
 - Anomaly detection using a GNN-based framework on a customized dataset
 - (Published) *Adaptive Sparsified Graph Learning Framework for Vessel Behavior Anomalies*
 - (Published) *STGVAD: Spatio-Temporal Graph-based Vessel Behavior Anomaly Detection*
- Visualization for Ship Collision Risk Assessment Mar 2024 – May 2025
 - Development of a decision supportive interactive system for ship collision avoidance
 - (Published) *FURIOUS: Fully unified risk-assessment with interactive operational user system for vessels*

VAIV Company

- Generative AI Professional Development Project Mar 2024 – Present
 - Task planning with LLM-guided GNNs

Seoul National University

- SNU Creative Education Project (Inno-Edu 2031) Jan 2024 – Dec 2024
 - Curriculum reform initiative under the SNU Creative Education Project

	- Undergraduate Course Development (L0655.005100) Veritas Practice: Data Visualization and Me	Jan 2024 – Jun 2024
AWARDS & SCHOLARSHIPS	SNU Medical AI Scholarship , Seoul National University	Aug 2024 – Present
	BK 21 Scholarship , Brain Korea	Mar 2025 – Present
	Veritas Scholarship , Korea University	Feb 2022
	- Research on detection of anomalies in the manufacturing process - Advisor: Prof. Sungwon Han	
	HAICON2021 , Korea Institute of Information Security & Cryptology (KIISC)	Nov 2021
	- Construction of an AI model for identifying time-series based anomalies in industrial control systems - Ranked 6th out of 177 teams	
	Industrial Engineering Project Competition , Korean Institute of Industrial Engineers (KIIE)	Oct 2021
	- Design of a dynamic corridor network for K-UAM - Received participation award in industrial engineering project competition	
	Data Visualization Competition , Korea University, Yonsei University, Intel	Aug 2021
	- Visualization of patient and their deductible premium relationship - Received grand prize	
PROFESSIONAL SERVICES	Conference Review	
	Learning on Graphs Conference (LOG)	2025
	Workshop Review	
	The Reach and Limits of AI for Scientific Discovery (AI4Science)@NeurIPS 2025	2025
	Journal Review	
	IEEE Access	2025
ACTIVITIES	Lab Captain , Visualization and Business Analytics Lab (ViBA)	Dec 2024 – Present
	Advisor: Prof. Hyunwoo Park	
	Samsung Dream Class , Samsung Welfare Foundation	Sep 2021 – Feb 2022
	Korea-China Leadership Program , Korea Foundation for Advanced Studies (KFAS)	Jul 2019
TEACHING EXPERIENCE	Teaching Assistant	
	K-Digital training program at Seoul National University	Jun 2024
	Python data analysis training program for Seoul National University faculty members	Jan 2024
TALKS AND SEMINARS	FURIOUS: Fully unified risk-assessment with interactive operational user system for vessels	
	2024 INFORMS Annual Meeting (Contributed Session)	Oct 2024
	Spatio-Temporal Dynamics-Aware GNN Framework for Vessel Behavior Anomaly Detection	
	2025 KDMS Summer (Best Paper Award)	Aug 2025
REFERENCES	Prof. Hyunwoo Park , Associate Professor, Seoul National University	
	Email: hyunwoopark@snu.ac.kr	

[CV compiled on 2025-09-11]